

In Problems 1–12, use the Laplace transform to solve the given differential equation subject to the indicated initial conditions.

1. $y' - 3y = \delta(t - 2)$, $y(0) = 0$
2. $y' + y = \delta(t - 1)$, $y(0) = 2$
3. $y'' + y = \delta(t - 2\pi)$, $y(0) = 0$, $y'(0) = 1$
4. $y'' + 16y = \delta(t - 2\pi)$, $y(0) = 0$, $y'(0) = 0$
5. $y'' + y = \delta(t - \pi/2) + \delta(t - 3\pi/2)$, $y(0) = 0$, $y'(0) = 0$
6. $y'' + y = \delta(t - 2\pi) + \delta(t - 4\pi)$, $y(0) = 1$, $y'(0) = 0$
7. $y'' + 2y' = \delta(t - 1)$, $y(0) = 0$, $y'(0) = 1$
8. $y'' - 2y' = 1 + \delta(t - 2)$, $y(0) = 0$, $y'(0) = 1$
9. $y'' + 4y' + 5y = \delta(t - 2\pi)$, $y(0) = 0$, $y'(0) = 0$
10. $y'' + 2y' + y = \delta(t - 1)$, $y(0) = 0$, $y'(0) = 0$
11. $y'' + 4y' + 13y = \delta(t - \pi) + \delta(t - 3\pi)$, $y(0) = 1$, $y'(0) = 0$
12. $y'' - 7y' + 6y = e^t + \delta(t - 2) + \delta(t - 4)$, $y(0) = 0$, $y'(0) = 0$

In Problems 13 and 14, use the Laplace transform to solve the given initial-value problem. Graph your solution on the interval $[0, 8\pi]$.

13. $y'' + y = \sum_{k=1}^{\infty} \delta(t - k\pi)$, $y(0) = 0$, $y'(0) = 1$
14. $y'' + y = \sum_{k=1}^{\infty} \delta(t - 2k\pi)$, $y(0) = 0$, $y'(0) = 1$

In Problems 15 and 16, a uniform beam of length L carries a concentrated load w_0 at $x = \frac{1}{2}L$. Solve the differential equation

$$EI \frac{d^4 y}{dx^4} = w_0 \delta(x - \tfrac{1}{2}L), \quad 0 < x < L,$$

subject to the given boundary conditions.

15. $y(0) = 0$, $y'(0) = 0$, $y''(L) = 0$, $y'''(L) = 0$

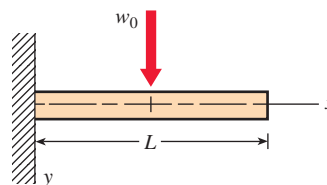


FIGURE 4.5.4 Beam embedded at its left end and free at its right end

16. $y(0) = 0$, $y'(0) = 0$, $y(L) = 0$, $y'(L) = 0$

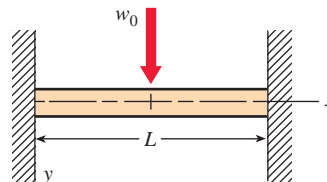


FIGURE 4.5.5 Beam embedded at both ends

Discussion Problems

17. Someone tells you that the solutions of the two IVPs

$$y'' + 2y' + 10y = 0, \quad y(0) = 0, \quad y'(0) = 1$$

and

$$y'' + 2y' + 10y = \delta(t), \quad y(0) = 0, \quad y'(0) = 0$$

are exactly the same. Do you agree or disagree? Defend your answer.

18. Reread (i) in the *Remarks* at the end of this section. Then use the Laplace transform to solve the initial-value problem:

$$y'' + 4y' + 3y = e^t \delta(t - 1), \quad y(0) = 0, \quad y'(0) = 2.$$

Use a graphing utility to graph $y(t)$ for $0 \leq t \leq 5$.

INTRODUCTION When initial conditions are specified, the Laplace transform reduces a system of linear differential equations with constant coefficients to a set of simultaneous algebraic equations in the transformed functions.

Coupled Springs In our first example we solve the model

$$\begin{aligned} m_1 x_1'' &= -k_1 x_1 + k_2 (x_2 - x_1) \\ m_2 x_2'' &= -k_2 (x_2 - x_1) \end{aligned} \quad (1)$$

that describes the motion of two masses m_1 and m_2 in the coupled spring/mass system shown in Figure 3.12.1 of Section 3.12.

EXAMPLE 1

Example 4 of Section 3.12 Revisited

Use the Laplace transform to solve

$$\begin{aligned} x_1'' + 10x_1 - 4x_2 &= 0 \\ -4x_1 + x_2'' + 4x_2 &= 0 \end{aligned} \quad (2)$$

subject to $x_1(0) = 0$, $x_1'(0) = 1$, $x_2(0) = 0$, $x_2'(0) = -1$. This is system (1) with $k_1 = 6$, $k_2 = 4$, $m_1 = 1$, and $m_2 = 1$.

SOLUTION The transform of each equation is

$$s^2 X_1(s) - s x_1(0) - x_1'(0) + 10 X_1(s) - 4 X_2(s) = 0$$

$$-4 X_1(s) + s^2 X_2(s) - s x_2(0) - x_2'(0) + 4 X_2(s) = 0,$$

where $X_1(s) = \mathcal{L}\{x_1(t)\}$ and $X_2(s) = \mathcal{L}\{x_2(t)\}$. The preceding system is the same as

$$\begin{aligned} (s^2 + 10)X_1(s) - 4X_2(s) &= 1 \\ -4X_1(s) + (s^2 + 4)X_2(s) &= -1. \end{aligned} \quad (3)$$

Solving (3) for $X_1(s)$ and using partial fractions on the result yields

$$X_1(s) = \frac{s^2}{(s^2 + 2)(s^2 + 12)} = -\frac{1/5}{s^2 + 2} + \frac{6/5}{s^2 + 12},$$

and therefore

$$\begin{aligned} x_1(t) &= -\frac{1}{5\sqrt{2}} \mathcal{L}^{-1}\left\{\frac{\sqrt{2}}{s^2 + 2}\right\} + \frac{6}{5\sqrt{12}} \mathcal{L}^{-1}\left\{\frac{\sqrt{12}}{s^2 + 12}\right\} \\ &= -\frac{\sqrt{2}}{10} \sin \sqrt{2}t + \frac{\sqrt{3}}{5} \sin 2\sqrt{3}t. \end{aligned}$$

Substituting the expression for $X_1(s)$ into the first equation of (3) gives us

$$X_2(s) = -\frac{s^2 + 6}{(s^2 + 2)(s^2 + 12)} = -\frac{2/5}{s^2 + 2} - \frac{3/5}{s^2 + 12}$$

and

$$\begin{aligned} x_2(t) &= -\frac{2}{5\sqrt{2}} \mathcal{L}^{-1}\left\{\frac{\sqrt{2}}{s^2 + 2}\right\} - \frac{3}{5\sqrt{12}} \mathcal{L}^{-1}\left\{\frac{\sqrt{12}}{s^2 + 12}\right\} \\ &= -\frac{\sqrt{2}}{5} \sin \sqrt{2}t - \frac{\sqrt{3}}{10} \sin 2\sqrt{3}t. \end{aligned}$$

Finally, the solution to the given system (2) is

$$\begin{aligned} x_1(t) &= -\frac{\sqrt{2}}{10} \sin \sqrt{2}t + \frac{\sqrt{3}}{5} \sin 2\sqrt{3}t \\ x_2(t) &= -\frac{\sqrt{2}}{5} \sin \sqrt{2}t - \frac{\sqrt{3}}{10} \sin 2\sqrt{3}t. \end{aligned} \quad (4)$$

The solution (4) is the same as (14) of Section 3.12. ≡

Networks In (18) of Section 2.9 we saw that currents $i_1(t)$ and $i_2(t)$ in the network containing an inductor, a resistor, and a capacitor shown in **FIGURE 4.6.1** were governed by the system of first-order differential equations

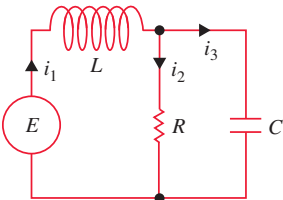


FIGURE 4.6.1 Electrical network

$$\begin{aligned} L \frac{di_1}{dt} + R i_2 &= E(t) \\ RC \frac{di_2}{dt} + i_2 - i_1 &= 0. \end{aligned} \quad (5)$$

We solve this system by the Laplace transform in the next example.

EXAMPLE 2

An Electrical Network

Solve the system in (5) under the conditions $E(t) = 60$ V, $L = 1$ h, $R = 50$ Ω , $C = 10^{-4}$ f, and the currents i_1 and i_2 are initially zero.

SOLUTION We must solve

$$\frac{di_1}{dt} + 50i_2 = 60$$

$$50(10^{-4}) \frac{di_2}{dt} + i_2 - i_1 = 0$$

subject to $i_1(0) = 0$, $i_2(0) = 0$.

Applying the Laplace transform to each equation of the system and simplifying gives

$$sI_1(s) + 50I_2(s) = \frac{60}{s}$$

$$-200I_1(s) + (s + 200)I_2(s) = 0,$$

where $I_1(s) = \mathcal{L}\{i_1(t)\}$ and $I_2(s) = \mathcal{L}\{i_2(t)\}$. Solving the system for I_1 and I_2 and decomposing the results into partial fractions gives

$$I_1(s) = \frac{60s + 12,000}{s(s + 100)^2} = \frac{6/5}{s} - \frac{6/5}{s + 100} - \frac{60}{(s + 100)^2}$$

$$I_2(s) = \frac{12,000}{s(s + 100)^2} = \frac{6/5}{s} - \frac{6/5}{s + 100} - \frac{120}{(s + 100)^2}.$$

Taking the inverse Laplace transform, we find the currents to be

$$i_1(t) = \frac{6}{5} - \frac{6}{5}e^{-100t} - 60te^{-100t}$$

$$i_2(t) = \frac{6}{5} - \frac{6}{5}e^{-100t} - 120te^{-100t}.$$

Note that both $i_1(t)$ and $i_2(t)$ in Example 2 tend toward the value $E/R = \frac{6}{5}$ as $t \rightarrow \infty$. Furthermore, since the current through the capacitor is $i_3(t) = i_1(t) - i_2(t) = 60te^{-100t}$, we observe that $i_3(t) \rightarrow 0$ as $t \rightarrow \infty$.

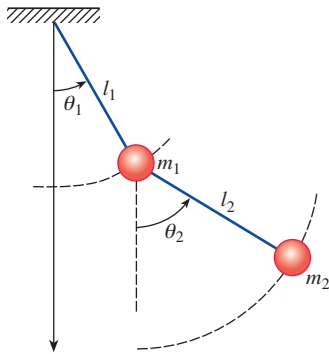


FIGURE 4.6.2 Double pendulum

Double Pendulum As shown in FIGURE 4.6.2, a double pendulum oscillates in a vertical plane under the influence of gravity. For small displacements $\theta_1(t)$ and $\theta_2(t)$, it can be shown that the system of differential equations describing the motion is

$$(m_1 + m_2)l_1^2\theta_1'' + m_2l_1l_2\theta_2'' + (m_1 + m_2)l_1g\theta_1 = 0 \quad (6)$$

$$m_2l_2^2\theta_2'' + m_2l_1l_2\theta_1'' + m_2l_2g\theta_2 = 0.$$

As indicated in Figure 4.6.2, θ_1 is measured (in radians) from a vertical line extending downward from the pivot of the system and θ_2 is measured from a vertical line extending downward from the center of mass m_1 . The positive direction is to the right and the negative direction is to the left.

EXAMPLE 3

Double Pendulum

It is left as an exercise to fill in the details of using the Laplace transform to solve system (6) when $m_1 = 3$, $m_2 = 1$, $l_1 = l_2 = 16$, $\theta_1(0) = 1$, $\theta_2(0) = -1$, $\theta_1'(0) = 0$, and $\theta_2'(0) = 0$. You should find that

$$\theta_1(t) = \frac{1}{4} \cos \frac{2}{\sqrt{3}}t + \frac{3}{4} \cos 2t \quad (7)$$

$$\theta_2(t) = \frac{1}{2} \cos \frac{2}{\sqrt{3}}t - \frac{3}{2} \cos 2t.$$

With the aid of a CAS, the positions of the two masses at $t = 0$ and at subsequent times are shown in **FIGURE 4.6.3**. See Problem 23 in Exercises 4.6.

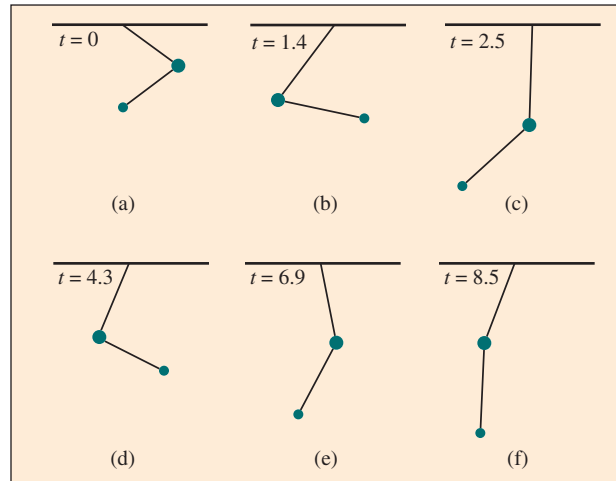


FIGURE 4.6.3 Positions of masses at various times in Example 3

4.6 Exercises

Answers to selected odd-numbered problems begin on page ANS-10.

In Problems 1–12, use the Laplace transform to solve the given system of differential equations.

1. $\frac{dx}{dt} = -x + y$

$$\frac{dy}{dt} = 2x$$

$$x(0) = 0, y(0) = 1$$

3. $\frac{dx}{dt} = x - 2y$

$$\frac{dy}{dt} = 5x - y$$

$$x(0) = -1, y(0) = 2$$

5. $2\frac{dx}{dt} + \frac{dy}{dt} - 2x = 1$

$$\frac{dx}{dt} + \frac{dy}{dt} - 3x - 3y = 2$$

$$x(0) = 0, y(0) = 0$$

7. $\frac{d^2x}{dt^2} + x - y = 0$

$$\frac{d^2x}{dt^2} + y - x = 0$$

$$x(0) = 0, x'(0) = -2, \\ y(0) = 0, y'(0) = 1$$

9. $\frac{d^2x}{dt^2} + \frac{d^2y}{dt^2} = t^2$

$$\frac{d^2x}{dt^2} - \frac{d^2y}{dt^2} = 4t$$

$$x(0) = 8, x'(0) = 0, \\ y(0) = 0, y'(0) = 0$$

2. $\frac{dx}{dt} = 2y + e^t$

$$\frac{dy}{dt} = 8x - t$$

$$x(0) = 1, y(0) = 1$$

4. $\frac{dx}{dt} + 3x + \frac{dy}{dt} = 1$

$$\frac{dx}{dt} - x + \frac{dy}{dt} - y = e^t$$

$$x(0) = 0, y(0) = 0$$

6. $\frac{dx}{dt} + x - \frac{dy}{dt} + y = 0$

$$\frac{dx}{dt} + \frac{dy}{dt} + 2y = 0$$

$$x(0) = 0, y(0) = 1$$

8. $\frac{d^2x}{dt^2} + \frac{dx}{dt} + \frac{dy}{dt} = 0$

$$\frac{d^2y}{dt^2} + \frac{dy}{dt} - 4\frac{dx}{dt} = 0$$

$$x(0) = 1, x'(0) = 0, \\ y(0) = -1, y'(0) = 5$$

10. $\frac{dx}{dt} - 4x + \frac{d^3y}{dt^3} = 6 \sin t$

$$\frac{dx}{dt} + 2x - 2\frac{d^3y}{dt^3} = 0$$

$$x(0) = 0, y(0) = 0, \\ y'(0) = 0, y''(0) = 0$$

11. $\frac{d^2x}{dt^2} + 3\frac{dy}{dt} + 3y = 0$

$$\frac{d^2x}{dt^2} + 3y = te^{-t}$$

$$x(0) = 0, x'(0) = 2, \\ y(0) = 0$$

12. $\frac{dx}{dt} = 4x - 2y + 2\mathcal{U}(t - 1)$

$$\frac{dy}{dt} = 3x - y + \mathcal{U}(t - 1)$$

$$x(0) = 0, y(0) = \frac{1}{2}$$

13. Solve system (1) when $k_1 = 3, k_2 = 2, m_1 = 1, m_2 = 1$ and $x_1(0) = 0, x'_1(0) = 1, x_2(0) = 1, x'_2(0) = 0$.

14. Derive the system of differential equations describing the straight-line vertical motion of the coupled springs shown in equilibrium in **FIGURE 4.6.4**. Use the Laplace transform to solve the system when $k_1 = 1, k_2 = 1, k_3 = 1, m_1 = 1, m_2 = 1$ and $x_1(0) = 0, x'_1(0) = -1, x_2(0) = 0, x'_2(0) = 1$.

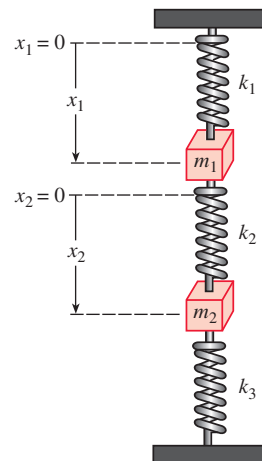


FIGURE 4.6.4 Coupled springs in Problem 14

15. (a) Show that the system of differential equations for the currents $i_2(t)$ and $i_3(t)$ in the electrical network shown in **FIGURE 4.6.5** is

$$L_1 \frac{di_2}{dt} + Ri_2 + Ri_3 = E(t)$$

$$L_2 \frac{di_3}{dt} + Ri_2 + Ri_3 = E(t).$$

- (b) Solve the system in part (a) if $R = 5 \Omega$, $L_1 = 0.01$ h, $L_2 = 0.0125$ h, $E = 100$ V, $i_2(0) = 0$, and $i_3(0) = 0$.

- (c) Determine the current $i_1(t)$.

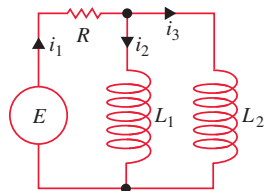


FIGURE 4.6.5 Network in Problem 15

16. (a) In Problem 14 in Exercises 2.9 you were asked to show that the currents $i_2(t)$ and $i_3(t)$ in the electrical network shown in **FIGURE 4.6.6** satisfy

$$L \frac{di_2}{dt} + L \frac{di_3}{dt} + R_1 i_2 = E(t)$$

$$-R_1 \frac{di_2}{dt} + R_2 \frac{di_3}{dt} + \frac{1}{C} i_3 = 0.$$

Solve the system if $R_1 = 10 \Omega$, $R_2 = 5 \Omega$, $L = 1$ h, $C = 0.2$ f,

$$E(t) = \begin{cases} 120, & 0 \leq t < 2 \\ 0, & t \geq 2, \end{cases}$$

$i_2(0) = 0$, and $i_3(0) = 0$.

- (b) Determine the current $i_1(t)$.

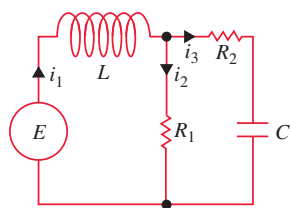


FIGURE 4.6.6 Network in Problem 16

17. Solve the system given in (17) of Section 2.9 when $R_1 = 6 \Omega$, $R_2 = 5 \Omega$, $L_1 = 1$ h, $L_2 = 1$ h, $E(t) = 50 \sin t$ V, $i_2(0) = 0$, and $i_3(0) = 0$.
18. Solve (5) when $E = 60$ V, $L = \frac{1}{2}$ h, $R = 50 \Omega$, $C = 10^{-4}$ f, $i_1(0) = 0$, and $i_2(0) = 0$.
19. Solve (5) when $E = 60$ V, $L = 2$ h, $R = 50 \Omega$, $C = 10^{-4}$ f, $i_1(0) = 0$, and $i_2(0) = 0$.

20. (a) Show that the system of differential equations for the charge on the capacitor $q(t)$ and the current $i_3(t)$ in the electrical network shown in **FIGURE 4.6.7** is

$$R_1 \frac{dq}{dt} + \frac{1}{C} q + R_1 i_3 = E(t)$$

$$L \frac{di_3}{dt} + R_2 i_3 - \frac{1}{C} q = 0.$$

- (b) Find the charge on the capacitor when $L = 1$ h, $R_1 = 1 \Omega$, $R_2 = 1 \Omega$, $C = 1$ f,

$$E(t) = \begin{cases} 0, & 0 < t < 1 \\ 50e^{-t}, & t \geq 1, \end{cases}$$

$i_3(0) = 0$, and $q(0) = 0$.

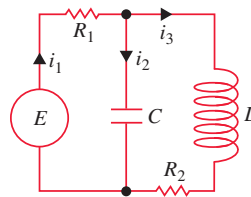


FIGURE 4.6.7 Network in Problem 20

Mathematical Models

21. **Range of a Projectile—No Air Resistance** If you worked Problem 23 in Exercises 3.12, you saw that when air resistance and all other forces except its weight $w = mg$ are ignored, the path of motion of a ballistic projectile, such as a cannon shell, is described by the system of linear differential equations

$$\begin{aligned} m \frac{d^2 x}{dt^2} &= 0 \\ m \frac{d^2 y}{dt^2} &= -mg. \end{aligned} \quad (8)$$

- (a) If the projectile is launched from level ground with an initial velocity $\mathbf{v}_0$ assumed to be tangent to its path of motion or trajectory, then the initial conditions accompanying the system are $x(0) = 0$, $x'(0) = v_0 \cos \theta$, $y(0) = 0$, $y'(0) = v_0 \sin \theta$ where $v_0 = \|\mathbf{v}_0\|$ the initial speed is constant and θ is the constant angle of elevation of the cannon. See Figure 3.R.5 in *The Paris Guns* problem on page 206. Use the Laplace transform to solve system (8).
- (b) The solutions $x(t)$ and $y(t)$ of the system in part (a) are parametric equations of the trajectory of the projectile. By using $x(t)$ to eliminate the parameter t in $y(t)$ show that the trajectory is parabolic.
- (c) Use the results of part (b) to show that the horizontal range R of the projectile is given by

$$R = \frac{v_0^2}{g} \sin 2\theta. \quad (9)$$

From (9) we not only see that R is a maximum when $\theta = \pi/4$ but that a projectile launched at distinct complementary angles θ and $\pi/2 - \theta$ has the same submaximum range. See **FIGURE 4.6.8**. Use a trigonometric identity to prove this last result.

- (d) Show that the maximum height H of the projectile is given by

$$H = \frac{v_0^2}{2g} \sin^2 \theta. \quad (10)$$

- (e) Suppose $g = 32 \text{ ft/s}^2$, $\theta = 38^\circ$, and $v_0 = 300 \text{ ft/s}$. Use (9) and (10) to find the horizontal range and maximum height of the projectile. Repeat with $\theta = 52^\circ$ and $v_0 = 300 \text{ ft/s}$.
- (f) Because formulas (9) and (10) are not valid in all cases (see Problem 22), it is advantageous to you to remember that the range and maximum height of a ballistic projectile can be obtained by working directly with $x(t)$ and $y(t)$, that is, by solving $y(t) = 0$ and $y'(t) = 0$. The first equation gives the time when the projectile hits the ground and the second gives the time when $y(t)$ is a maximum. Find these times and verify the range and maximum height obtained in part (e) for the trajectory with $\theta = 38^\circ$ and $v_0 = 300 \text{ ft/s}$. Repeat with $\theta = 52^\circ$.
- (g) With $g = 32 \text{ ft/s}^2$, $\theta = 38^\circ$ and $v_0 = 300 \text{ ft/s}$ use a graphing utility or CAS to plot the trajectory of the projectile defined by the parametric equations $x(t)$ and $y(t)$ in part (a). Repeat with $\theta = 52^\circ$. Using different colors superimpose both curves on the same coordinate system.

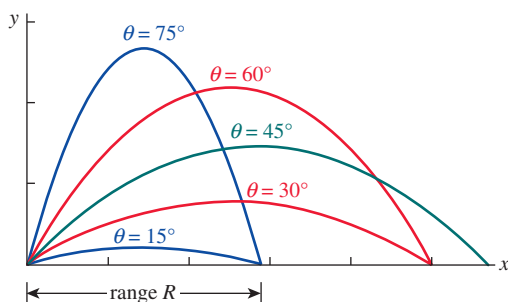


FIGURE 4.6.8 Projectiles in Problem 21

22. Range of a Projectile—With Linear Air Resistance In *The Paris Guns* problem on pages 206–207, the effect that nonlinear air resistance has on the trajectory of a cannon shell was examined numerically. In this problem we consider linear air resistance on a projectile.

- (a) Suppose that air resistance is a retarding force tangent to the path of the projectile but acts opposite to the motion. If we take air resistance to be proportional to the velocity of the projectile, then from Problem 24 of Exercises 3.12 the motion of the projectile is described by the system of linear differential equations

$$\begin{aligned} m \frac{d^2 x}{dt^2} &= -\beta \frac{dx}{dt} \\ m \frac{d^2 y}{dt^2} &= -mg - \beta \frac{dy}{dt}, \end{aligned} \quad (11)$$

where $\beta > 0$ is a constant. Use the Laplace transform to solve system (11) subject to the initial conditions in part (a) of Problem 21.

- (b) Suppose $m = \frac{1}{4} \text{ slug}$, $g = 32 \text{ ft/s}^2$, $\beta = 0.02$, $\theta = 38^\circ$, and $v_0 = 300 \text{ ft/s}$. Use a calculator or CAS to approximate the time when the projectile hits the ground and then compute $x(t)$ to find its corresponding horizontal range.
- (c) The complementary-angle property in part (c) of Problem 21 does not hold when air resistance is taken into consideration. To show this, repeat part (b) using the complementary angle $\theta = 52^\circ$ and compare the horizontal range with that found in part (b).
- (d) With $m = \frac{1}{4} \text{ slug}$, $g = 32 \text{ ft/s}^2$, $\beta = 0.02$, $\theta = 38^\circ$, and $v_0 = 300 \text{ ft/s}$, use a graphing utility or CAS to plot the trajectory of the projectile defined by the parametric equations $x(t)$ and $y(t)$. Repeat with $\theta = 52^\circ$. Using different colors, superimpose both of these curves along with the two curves in part (g) of Problem 21 on the same coordinate system.

Computer Lab Assignment

- 23. (a)** Use the Laplace transform and the information given in Example 3 to obtain the solution (7) of the system given in (6).
- (b)** Use a graphing utility to plot the graphs of $\theta_1(t)$ and $\theta_2(t)$ in the $t\theta$ -plane. Which mass has extreme displacements of greater magnitude? Use the graphs to estimate the first time that each mass passes through its equilibrium position. Discuss whether the motion of the pendulums is periodic.
- (c)** As parametric equations, graph $\theta_1(t)$ and $\theta_2(t)$ in the $\theta_1\theta_2$ -plane. The curve defined by these parametric equations is called a **Lissajous curve**.
- (d)** The position of the masses at $t = 0$ is given in Figure 4.6.3(a). Note that we have used $1 \text{ radian} \approx 57.3^\circ$. Use a calculator or a table application in a CAS to construct a table of values of the angles θ_1 and θ_2 for $t = 1, 2, \dots, 10$ seconds. Then plot the positions of the two masses at these times.
- (e)** Use a CAS to find the first time that $\theta_1(t) = \theta_2(t)$ and compute the corresponding angular value. Plot the positions of the two masses at these times.
- (f)** Utilize a CAS to also draw appropriate lines to simulate the pendulum rods as in Figure 4.6.3. Use the animation capability of your CAS to make a “movie” of the motion of the double pendulum from $t = 0$ to $t = 10$ using a time increment of 0.1. [Hint: Express the coordinates $(x_1(t), y_1(t))$ and $(x_2(t), y_2(t))$ of the masses m_1 and m_2 , respectively, in terms of $\theta_1(t)$ and $\theta_2(t)$.]

In Problems 1 and 2, use the definition of the Laplace transform to find $\mathcal{L}\{f(t)\}$.

$$1. f(t) = \begin{cases} t, & 0 \leq t < 1 \\ 2 - t, & t \geq 1 \end{cases} \quad 2. f(t) = \begin{cases} 0, & 0 \leq t < 2 \\ 1, & 2 \leq t < 4 \\ 0, & t \geq 4 \end{cases}$$

In Problems 3–24, fill in the blanks or answer true/false.

3. If f is not piecewise continuous on $[0, \infty)$, then $\mathcal{L}\{f(t)\}$ will not exist. _____
4. The function $f(t) = (e^t)^{10}$ is not of exponential order. _____
5. $F(s) = s^2/(s^2 + 4)$ is not the Laplace transform of a function that is piecewise continuous and of exponential order. _____
6. If $\mathcal{L}\{f(t)\} = F(s)$ and $\mathcal{L}\{g(t)\} = G(s)$, then $\mathcal{L}^{-1}\{F(s)G(s)\} = f(t)g(t)$. _____
7. $\mathcal{L}\{e^{-7t}\} =$ _____
8. $\mathcal{L}\{te^{-7t}\} =$ _____
9. $\mathcal{L}\{\sin 2t\} =$ _____
10. $\mathcal{L}\{e^{-3t} \sin 2t\} =$ _____
11. $\mathcal{L}\{t \sin 2t\} =$ _____
12. $\mathcal{L}\{\sin 2t \mathcal{U}(t - \pi)\} =$ _____
13. $\mathcal{L}^{-1}\left\{\frac{20}{s^6}\right\} =$ _____
14. $\mathcal{L}^{-1}\left\{\frac{1}{3s - 1}\right\} =$ _____
15. $\mathcal{L}^{-1}\left\{\frac{1}{(s - 5)^3}\right\} =$ _____
16. $\mathcal{L}^{-1}\left\{\frac{1}{s^2 - 5}\right\} =$ _____
17. $\mathcal{L}^{-1}\left\{\frac{s}{s^2 - 10s + 29}\right\} =$ _____
18. $\mathcal{L}^{-1}\left\{\frac{e^{-5s}}{s^2}\right\} =$ _____
19. $\mathcal{L}^{-1}\left\{\frac{s + \pi}{s^2 + \pi^2} e^{-s}\right\} =$ _____
20. $\mathcal{L}^{-1}\left\{\frac{1}{L^2 s^2 + n^2 \pi^2}\right\} =$ _____
21. $\mathcal{L}\{e^{-5t}\}$ exists for $s >$ _____.
22. If $\mathcal{L}\{f(t)\} = F(s)$, then $\mathcal{L}\{te^{8t}f(t)\} =$ _____.
23. If $\mathcal{L}\{f(t)\} = F(s)$ and $k > 0$, then $\mathcal{L}\{e^{at}f(t - k)\mathcal{U}(t - k)\} =$ _____.
24. $\mathcal{L}\left\{\int_0^t e^{a\tau}f(\tau)d\tau\right\} =$ _____ whereas $\mathcal{L}\left\{e^{at}\int_0^t f(\tau)d\tau\right\} =$ _____.

In Problems 25–28, use the unit step function to write down an equation for each graph in terms of the function $y = f(t)$ whose graph is given in **FIGURE 4.R.1**.

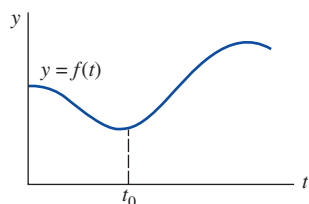


FIGURE 4.R.1 Graph for Problems 25–28

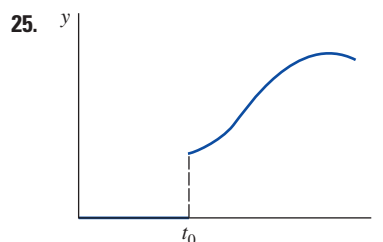


FIGURE 4.R.2 Graph for Problem 25

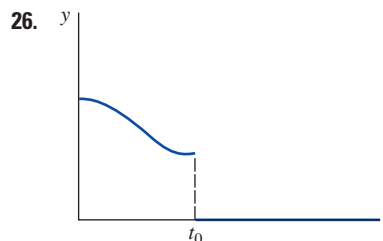


FIGURE 4.R.3 Graph for Problem 26

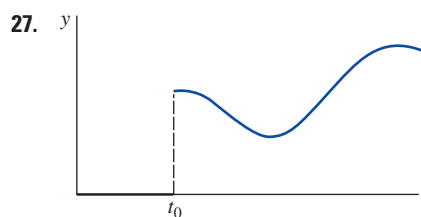


FIGURE 4.R.4 Graph for Problem 27

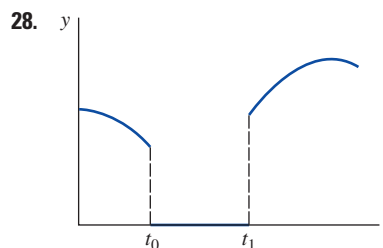


FIGURE 4.R.5 Graph for Problem 28

In Problems 29–32, express f in terms of unit step functions. Find $\mathcal{L}\{f(t)\}$ and $\mathcal{L}\{e^t f(t)\}$.

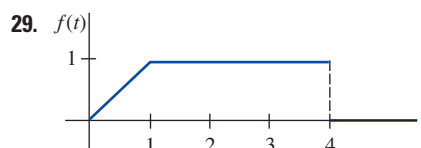


FIGURE 4.R.6 Graph for Problem 29